Bright Minds™ Resource Activity

Animal Cell Model

Background

Curious about what we're made of? Find out what's inside an animal cell in the following experiment and make your own model cell using Plaster of Paris, plasticine, water and paint.

Materials

- Plasticine modelling clay
- Plaster of Paris (available from hardware stores)
- Water
- Take-away food containers (you could use round for animal cells; rectangular or square for plant cells)
- Acrylic paints
- Paint brushes

What to do

1. Use the internet or text books to find out the size and shapes of cells and organelles. These are some representative values as a guide, but remember there are a wide range of cell and organelle sizes.
   Animal cell 20 – 50 µm
   Plant cell 100 – 150 µm
   Nucleus 2 – 10 µm
   Mitochondrion 2 – 10 µm x 1 – 2 µm
   Lysosome 2 µm
   Endoplasmic reticulum Up to 5 µm x 10 µm
   Golgi Apparatus 4 – 10 µm x 2 – 3 µm (4 - 7 membrane stacks high)
   Chloroplast 2 – 3 µm x 5 – 10 µm
   Plant vacuole Up to 50% of cell volume
   Centriole 0.4 µm
   Ribosome 0.02 µm

2. Calculate the magnification factor for the model, then calculate the size of the organelles.
   If the mould is 10 cm and you are modelling an animal cell of 50 µm, the magnification is 100,000/ 50 = 2000, so the nucleus should be 10 x 2,000 µm = 2 cm.

3. Use plasticine to make models of cellular organelles at the correct sizes. Make sure each model is less that 1.5 cm in height.
   Nucleus with nucleolus and nuclear pores
   Mitochondria
   Lysosomes
   Endoplasmic reticulum
   Golgi apparatus
   Chloroplasts (for plant cell)
   Vacuole (for plant cell)

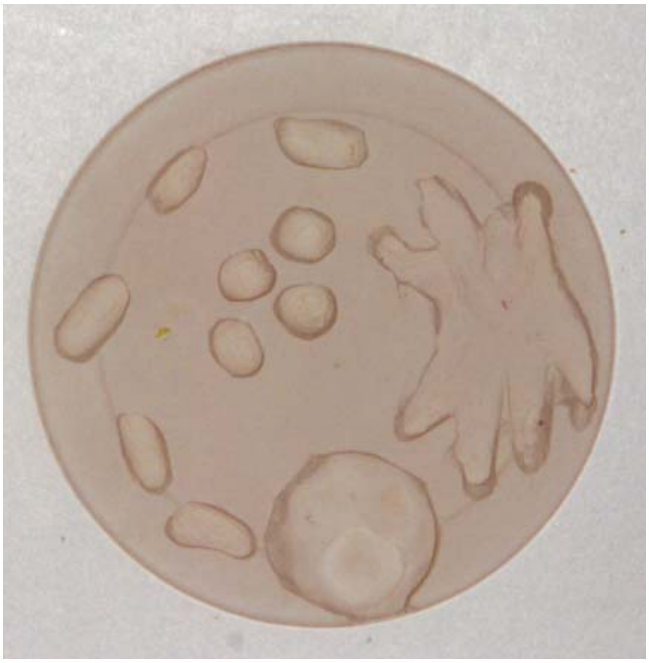


(a) plasticine moulds of cellular organelles in an animal cell, in a take-away food container;

4. Place the organelle models into the mould at appropriate places. Press down to flatten the underside. Note that the nucleus does not have to be in the centre of the cell.

What to do Cont.

5. Measure the amount of water required and sprinkle the weighed amount of Plaster of Paris powder over the water. The proportion should be 100g powder to 75mL of water. For a round container 10cm in diameter, use 100 – 150g; for a rectangular contained use 200 – 250 g. Stir until a thick homogenous slurry is formed.
6. Pour the slurry into the mould, making sure that each organelle is covered by at least 2mm. Add a picture hook to the back (ie top once the slurry is poured in) if you wish.
7. Allow the plaster to set to the stage where you can touch it without making a mark, but not to complete dryness. This will take between a few hours and overnight.
8. Tip the plaster model from the mould. If possible flex the container walls to make it easier to remove.



(b) plaster of Paris model before painting;

9. Remove the plasticine from the model. Use a toothpick, knife or skewer to dig the plasticine out. It should come away cleanly from the sides.
10. Trim any overhangs where each organelle model was removed. This will be easy if the model is still slightly damp.



(c) the painted model showing nucleus (green) with nucleolus (dark green) and nuclear pores (light green), mitochondria (yellow), lysosomes (orange), endoplasmic reticulum (blue) and ribosomes (dark blue);

11. Allow the model to dry more fully then paint with acrylic paints. A thin paint brush can be used to add fine detail which could not be modelled in plasticine. Once the paint is dry the model can be varnished and hung on the wall.

For Questions and Answers relating to this activity, and for other activities, useful teacher tips and science games for students, please check out the ***Bright Minds***™ website at ***www.brightminds.uq.edu.au***